Differential Equations

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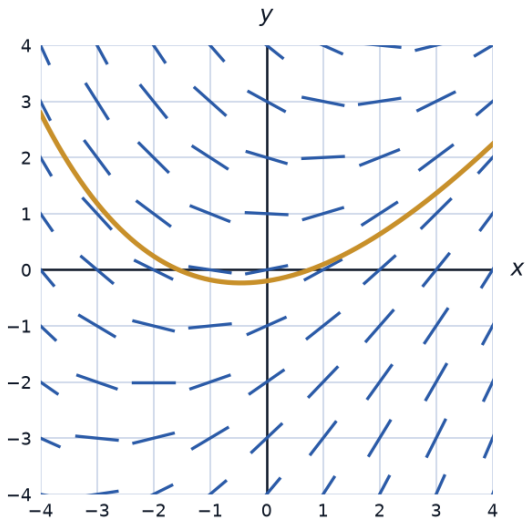
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Solving by separation of variables

Solve the differential equation $\frac{dy}{dx} = \frac{x}{y}$, given that $y = 3$ when $x = 0$.

2 [Foundational] • Target: Solving by separation of variables

Solve $\frac{dy}{dx} = 2xy$, given that $y = 5$ when $x = 0$.



3 [Foundational] • Target: Modeling with differential equations

A population P grows at a rate proportional to its current size, modeled by $\frac{dP}{dt} = kP$. If the population doubles every 5 years, find k, to three decimal places.

4 [Foundational] • Target: Modeling with differential equations

Newton's law of cooling states $\frac{dT}{dt} = -k(T - T_a)$, where T_a is ambient temperature. Solve this differential equation for $T(t)$ in terms of the initial temperature T_0.

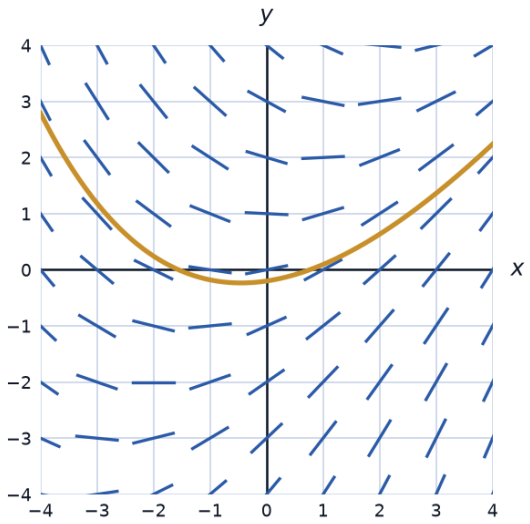
5 [Foundational] • Target: First-order linear differential equations

Solve $\frac{dy}{dx} + 2y = 6$ using an integrating factor, given $y = 1$ when $x = 0$.

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6 [Foundational] • Target: Euler's method for numerical solutions

Use Euler's method with step size $h = 0.5$ to estimate $y(1)$ for $\frac{dy}{dx} = x + y$, given $y(0) = 1$, showing each step.



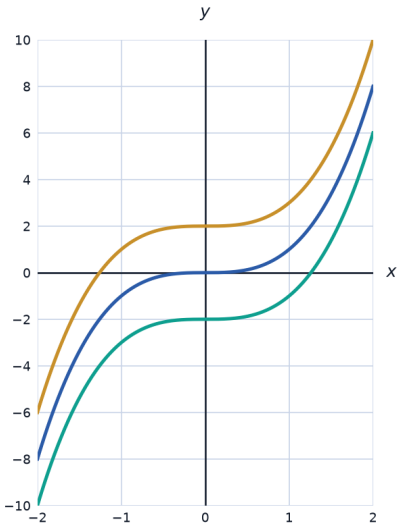
7 [Foundational] • Target: Second-order context: simple harmonic motion

The differential equation $\frac{d^2x}{dt^2} = -\omega^2 x$ describes simple harmonic motion. Verify that $x(t) = A\cos(\omega t)$ is a solution.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Family of solution curves and particular solutions

Find the general solution of $\frac{dy}{dx} = 3x^2$, and explain what the arbitrary constant represents geometrically.



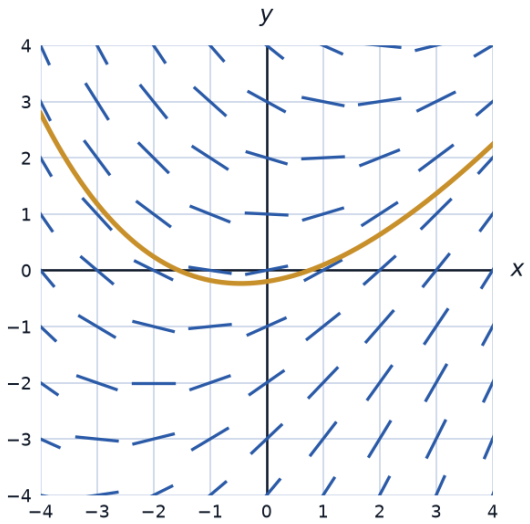
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9 [Intermediate] • Target: Family of solution curves and particular solutions

A tank drains such that $\frac{dV}{dt} = -k\sqrt{V}$ for some constant $k > 0$, where V is the volume of water. Solve this differential equation for $V(t)$, given $V(0) = V_0$.

10 [Intermediate] • Target: Verifying a solution to a second-order differential equation

Verify that $y = e^{2x}$ is a solution of the differential equation $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$.



11 [Intermediate] • Target: Slope fields and qualitative solutions

A differential equation is given by $\frac{dy}{dx} = x - y$. Find the gradient of the solution curve at the points $(0, 0)$, $(1, 0)$, and $(1, 1)$, and explain what a slope field represents.

12 [Intermediate] • Target: Solving a linear differential equation with an initial condition

Solve $\frac{dy}{dx} - \frac{y}{x} = x$ for $x > 0$ using an integrating factor, given $y = 2$ when $x = 1$.

13 [Intermediate] • Target: A logistic growth model

A population P grows according to the logistic differential equation $\frac{dP}{dt} = kP(10 - P)$. Using the partial fraction decomposition $\frac{1}{P(10-P)} = \frac{1}{10P} + \frac{1}{10(10-P)}$, solve for P in terms of t, given $P(0) = 1$ and leaving k as an unknown constant.

14 [Intermediate] • Target: Checking the order and degree of a differential equation

State the order of each of the following differential equations: (a) $\frac{dy}{dx} = 3x^2 - 2y$, (b) $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - y = 0$, (c) $\frac{d^3y}{dx^3} = \sin x$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

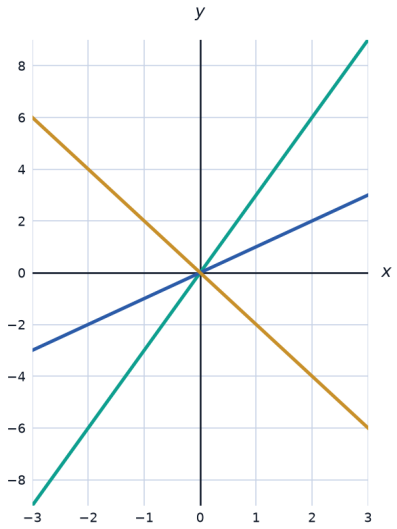
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15 [Challenging] • Target: Using Euler's method with a smaller step size • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Use Euler's method with step size $h = 0.25$ to estimate $y(0.5)$ for $\frac{dy}{dx} = 2x$, given $y(0) = 1$, showing each step, and compare with the exact solution $y = x^2 + 1$.

16 [Challenging] • Target: Using Euler's method with a smaller step size • Cross-Lesson: 4.02 The Poisson Distribution

Solve the differential equation $\frac{dy}{dx} = \frac{y}{x}$ for $x > 0$, given $y = 6$ when $x = 2$, and describe the family of curves this represents.



17 [Challenging] • Target: Using Euler's method with a smaller step size • Cross-Lesson: 5.01 Further Integration Techniques

A cup of coffee at $90\,^\circ$C is placed in a room at ambient temperature $20\,^\circ$C. Using Newton's law of cooling, $T(t) = T_a + (T_0 - T_a)e^{-kt}$, and given that after 10 minutes the coffee has cooled to $60\,^\circ$C, find the value of k to three decimal places.

18 [Challenging] • Target: Using Euler's method with a smaller step size • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

This question has three parts. A tank initially contains 200 litres of pure water. Salt water containing 3 grams of salt per litre flows in at a rate of 2 litres per minute, and the well-mixed solution flows out at the same rate, so the volume stays constant at 200 litres. Let $S(t)$ be the amount of salt (in grams) in the tank at time t minutes. (a) Explain why $\frac{dS}{dt} = 6 - \frac{S}{100}$. (b) Solve this differential equation for $S(t)$, given $S(0) = 0$. (c) Find the amount of salt in the tank after 50 minutes, to the nearest gram.

19 [Challenging] • Target: First-order linear differential equations • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

A cubic equation with real coefficients $x^3 + 7x^2 + 15x + 25 = 0$ has $-1 + 2i$ as one root. Find the real root.

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20 [Challenging] • Target: Euler's method for numerical solutions • Cross-Lesson: 6.01 Matrices (AI HL)

A cubic equation with real coefficients $x^3 - 10x^2 + 37x - 52 = 0$ has $3 + 2i$ as one root. Find the real root.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Separation of Variables

Solve the differential equation $\frac{dy}{dx} = \frac{2x}{y}$ with initial condition $y(0) = 4$ for $y > 0$.

a. $y = \sqrt{2x^2 + 16}$

b. $y = 2x^2 + 4$

c. $y = \sqrt{x^2 + 16}$

d. $y = x^2 + 4$

22 [Foundational] • Target: Separation of Variables

Solve $\frac{dy}{dx} = -3y$ with $y(0) = 50$.

a. $y = 50e^{-3x}$

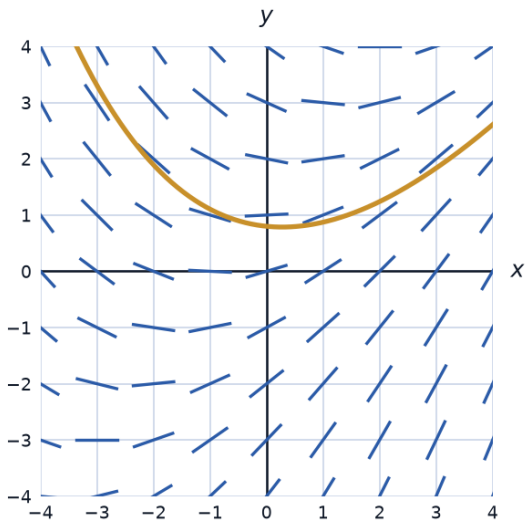
b. $y = 50e^{3x}$

c. $y = -3x + 50$

d. $y = 50 - 3e^x$

23 [Foundational] • Target: Separation of Variables

Solve $\frac{dy}{dx} = (1 + y^2)e^x$ with $y(0) = 0$.



a. $y = \tan(e^x - 1)$

b. $y = \arctan(e^x - 1)$

c. $y = \tan(e^x)$

d. $y = e^x - 1$

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24 [Foundational] • Target: Integrating Factor Method for First-Order Linear ODEs

For the linear differential equation $\frac{dy}{dx} + P(x)y = Q(x)$, what is the integrating factor $I(x)$?

a. $I(x) = e^{\int P(x)dx}$

b. $I(x) = \int P(x)dx$

c. $I(x) = e^{-\int P(x)dx}$

d. $I(x) = e^{\int Q(x)dx}$

25 [Foundational] • Target: Integrating Factor Method for First-Order Linear ODEs

Find the integrating factor for $\frac{dy}{dx} + \frac{3}{x}y = x^2$ for $x > 0$.

a. x^3

b. $3\ln x$

c. e^{3x}

d. $\frac{1}{x^3}$

26 [Foundational] • Target: Integrating Factor Method for First-Order Linear ODEs

Solve $\frac{dy}{dx} + 2y = 4$ with $y(0) = 5$.

a. $y = 2 + 3e^{-2x}$

b. $y = 2 + 5e^{-2x}$

c. $y = 4 + e^{-2x}$

d. $y = 2 - 3e^{-2x}$

27 [Foundational] • Target: Homogeneous Differential Equations (y = vx)

Which substitution transforms a homogeneous differential equation $\frac{dy}{dx} = f(y/x)$ into a separable equation?

a. $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

b. $y = v + x$

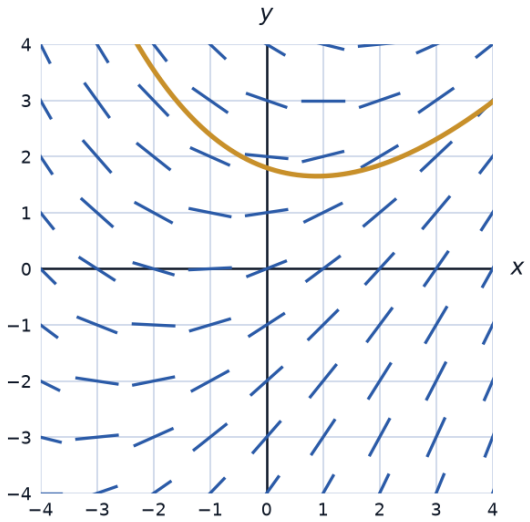
c. $v = xy$

d. $y = v^2x$

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28 [Foundational] • Target: Homogeneous Differential Equations (y = vx)

Classify the differential equation $\frac{dy}{dx} = \frac{x^2 + y^2}{2xy}$.



a. Homogeneous of degree 0

b. First-order linear

c. Exact non-linear

d. Second-order Bernoulli

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Homogeneous Differential Equations (y = vx)

Using $y = vx$, what does $\frac{dy}{dx} = \frac{y}{x} + 1$ become?

a. $x\frac{dv}{dx} = 1$

b. $v + \frac{dv}{dx} = 1$

c. $x\frac{dv}{dx} = v + 1$

d. $\frac{dv}{dx} = x$

30 [Intermediate] • Target: Numerical Euler's Method

What is Euler's recurrence relation for approximating solutions to $\frac{dy}{dx} = f(x, y)$ with step size h?

a. $y_{n+1} = y_n + hf(x_n, y_n)$

b. $y_{n+1} = y_n + \frac{h}{2}f(x_n, y_n)$

c. $y_{n+1} = f(x_n + h, y_n)$

d. $y_{n+1} = y_n + h^2 f(x_n, y_n)$

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31 [Intermediate] • Target: Numerical Euler's Method

Given $\frac{dy}{dx} = x + y$ with $y(0) = 1$, use Euler's method with step size $h = 0.1$ to find y_1 at $x = 0.1$.

a. 1.10

b. 1.05

c. 1.21

d. 1.15

32 [Intermediate] • Target: Numerical Euler's Method

Using the result above ($x_1 = 0.1, y_1 = 1.10$), find y_2 at $x = 0.2$ with $h = 0.1$.

a. 1.22

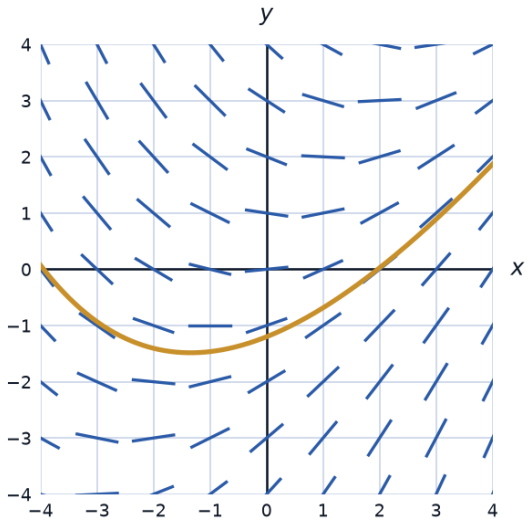
b. 1.20

c. 1.25

d. 1.31

33 [Intermediate] • Target: Separation of Variables

A cubic equation with real coefficients $x^3 + 6x^2 + 10x + 8 = 0$ has $-1 + i$ as one root. Find the real root.



a. -7

b. -1

c. -5

d. -4

34 [Intermediate] • Target: Integrating Factor Method for First-Order Linear ODEs

A cubic equation with real coefficients $x^3 - 8x^2 + 22x - 20 = 0$ has $3 + i$ as one root. Find the real root.

a. -2

b. 3

c. 2

d. 1

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35 [Intermediate] • Target: Homogeneous Differential Equations (y = vx)

A cubic equation with real coefficients $x^3 - 2x + 4 = 0$ has $1 + i$ as one root. Find the real root.

a. 2

b. 0

c. -6

d. -2

36 [Intermediate] • Target: Numerical Euler's Method

A graph has edges $1 - 2, 2 - 3, 3 - 4, 4 - 1$. Does it have an Euler circuit, an Euler trail (not circuit), or neither?

a. Euler circuit exists

b. Euler trail exists, but not an Euler circuit

c. Cannot be determined from the given information

d. Neither an Euler trail nor circuit exists

37 [Intermediate] • Target: Solving by separation of variables

A graph has edges $1 - 2, 2 - 3, 3 - 4, 4 - 1, 1 - 3$. Does it have an Euler circuit, an Euler trail (not circuit), or neither?

a. Neither an Euler trail nor circuit exists

b. Euler circuit exists

c. Euler trail exists, but not an Euler circuit

d. Cannot be determined from the given information

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Modeling with differential equations • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

A graph has edges $1-2, 2-3, 3-4, 1-4, 1-3, 2-4$. Does it have an Euler circuit, an Euler trail (not circuit), or neither?

a. Euler circuit exists

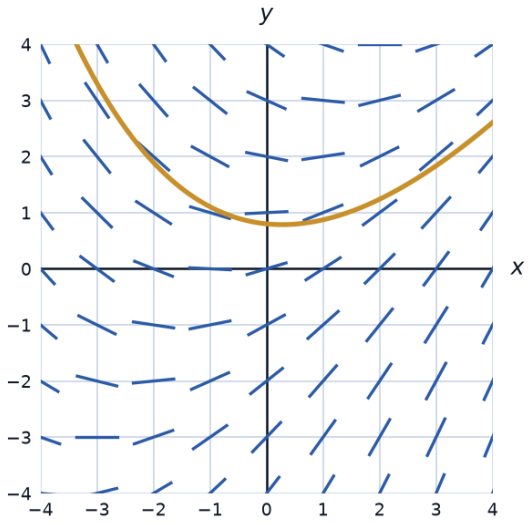
b. Neither an Euler trail nor circuit exists

c. Cannot be determined from the given information

d. Euler trail exists, but not an Euler circuit

39 [Challenging] • Target: First-order linear differential equations • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

Factor $P(x) = x^3 + 7x^2 + 14x + 8$ completely. Find the largest root.



a. -1

b. -2

c. -3

d. 0

40 [Challenging] • Target: Euler's method for numerical solutions • Cross-Lesson: 6.01 Matrices (AI HL)

Factor $P(x) = x^3 + 4x^2 - 9x - 36$ completely. Find the largest root.

a. 0

b. 3

c. 1

d. 2

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41 [Challenging] • Target: Second-order context: simple harmonic motion • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

Factor $P(x) = x^3 + 2x^2 - x - 2$ completely. Find the largest root.

a. -2

b. 4

c. 1

d. 2

42 [Challenging] • Target: Family of solution curves and particular solutions • Cross-Lesson: 6.03 Markov Chains (AI HL)

Factor $P(x) = x^3 + 2x^2 - 11x - 12$ completely. Find the largest root.

a. 3

b. 5

c. 6

d. 4

43 [Challenging] • Target: Verifying a solution to a second-order differential equation • Cross-Lesson: 1.01 Proof by Mathematical Induction

Factor $P(x) = x^3 - x^2 - 4x + 4$ completely. Find the largest root.

a. 2

b. 1

c. 4

d. 3

44 [Challenging] • Target: Slope fields and qualitative solutions • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

Factor $P(x) = x^3 - 2x^2 - 9x + 18$ completely. Find the largest root.

a. 0

b. 2

c. 3

d. 4

45 [Challenging] • Target: Solving a linear differential equation with an initial condition • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Factor $P(x) = x^3 - 4x^2 - 4x + 16$ completely. Find the largest root.

a. 3

b. 4

c. 5

d. 6

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